

YEAR 11

- 1) Determine the X and Y intercepts of this quadratic function.

$$Y = x^2 + 8x + 12$$

- 2) Complete the square of this $y = \frac{-1}{4}x^2 + 2x - 4$, then state the vertex.
- 3) Alex wants to see how high his slingshot can send an object into the air. He attaches his digital altimeter to a potato and then shoots the pair into the air. The potato's height in relation to time follows the pathway $h = 7t^2 + 21t + 5$, where h is height in feet and t is time in seconds. At what height was the potato at the exact moment it was released?
- 4) The CN tower in Toronto was once the world's tallest free-standing structure. From a certain point on the ground, the angle of elevation to the top of the tower is 65° . From a point 56.6m closer to the tower, the angle of elevation to the top of the tower is 70° . Calculate the height of the CN tower to the nearest metre.

Consider the sequence: $\frac{1}{2}; 4; \frac{1}{4}; 7; \frac{1}{8}; 10; \dots$ (Question 5-6)

- 5) If the pattern continues in the same way, write down the next two terms in the sequence.
- 6) Calculate the sum of the first 50 terms of the sequence.

Solve using the quadratic formula (Question 7-8)

- 7) $4x^2 - 12x + 9 = 0$
- 8) $8w^2 + 3w - 2 = 0$

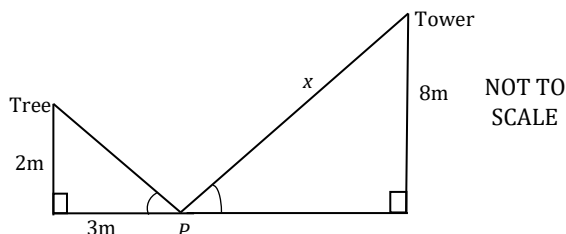
From a group of 15 hockey players at a game of hockey, 13 played on the field, 7 sat on the bench and 5 both played and sat on the bench. A hockey player is chosen at random from the team. Let A be the event the person played on the field and B be the event the person sat on the bench. (Question 9-12)

- 9) Represents the information in a two-way table.
- 10) Find the probability that the person only sat on the bench.
- 11) Find the probability that the person sat on the bench, given that they played on the field.
- 12) Find the probability that the person played on the field given that they sat on the bench.
- 13) Solve the following triangle:

$$\Delta WXY \text{ where } w = 120\text{m}, x = 77\text{m}, \text{ and } y = 115\text{m}$$

- 14) A point P lies between a tree, 2 meters high, and a tower, 8 meters high. P is 3 meters away from the base of the tree. From P , the angles of elevation to the top of the tree and to the top of the tower are equal.

What is the distance x , from P to the top of the tower?



Consider the sequence: 8; 18; 30; 44; ... (Question 15-16)

- 15) Calculate the n th term of the sequence.
16) Which term of the sequence are 330?

Given that geometric series: $8x^2 + 4x^2 + 2x^2 + \dots$ (Question 17-19)

- 17) Determine the n th term of the series.
18) For what value (s) of x will the series converge?
19) Calculate the sum of the series to infinity if $x = \frac{3}{2}$

The following information is given about the locations of three towns X , Y and Z :

- X is due east of Z
- X is on a bearing of 145° from Y
- Y is on a bearing of 060° from Z

- 20) Draw a diagram that best represents this information.